

**SIMATS ENGINEERING**  
**CO3 - ASSESSMENT TOOL 3**

**Comparative Analysis**

**COURSE NAME: DEEP LEARNING**

**COURSE CODE: MLA0408**

**COURSE OUTCOME COVERED**

**CO3:** Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Assessment Tool Weightage: 40%

**QUESTIONS:**

**Total Marks:25**

**Q1. CNN vs Fully Connected Neural Network**

A medical imaging organization wants to classify X-ray images using deep learning. Compare a **CNN with a traditional fully connected neural network** in terms of architectural design, motivation, spatial feature extraction, number of parameters, computational complexity, and ability to handle image data. Analyze the roles of convolutional layers, filters, and pooling layers, and justify why CNNs are more suitable for image classification applications.

**Q2. Different CNN Layers and Filters**

A manufacturing company is developing a CNN-based system to detect defects in industrial products. Compare the functions of **convolutional layers, pooling layers, activation layers, and fully connected layers** in terms of feature extraction, spatial information, computational requirements, and classification. Further compare **early-layer filters and deeper-layer filters** based on the type of features they learn, and justify how these layers collectively improve defect detection.

**Q3. Parameter Sharing vs Independent Weights**

A smart surveillance system needs to process thousands of high-resolution images efficiently. Compare **parameter sharing in CNNs with independent weights in fully connected networks** in terms of the number of parameters, memory requirements, computational complexity, feature detection, and translation-related invariance. Analyze how the reuse of convolutional filter weights improves learning efficiency and justify its importance for large-scale image-processing applications.

#### Q4. AlexNet vs ResNet

A computer vision company is developing an image classification system and is considering **AlexNet and ResNet** as possible architectures. Compare the two architectures in terms of network depth, layer organization, convolutional filters, parameter requirements, feature extraction capability, training difficulty, vanishing-gradient problem, computational cost, and classification performance. Recommend the more suitable architecture for a complex image recognition application and justify your choice.

#### Q5. Regularized CNN vs Non-Regularized CNN

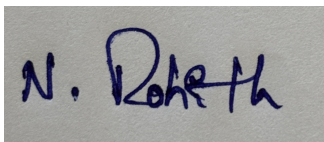
An agricultural organization is developing a CNN to classify plant diseases from leaf images. The initial model achieves very high training accuracy but performs poorly on unseen images. Compare a **CNN without regularization and a CNN using dropout, data augmentation, and batch normalization** in terms of overfitting, generalization, training stability, computational requirements, and model performance. Analyze which regularization techniques should be applied and justify the most appropriate approach for reliable real-world disease classification.

#### Code of Conduct:

I **NALIPI REDDY ROHITH/192425115**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

#### Signature of the student:

A rectangular box containing a handwritten signature in blue ink. The signature appears to be 'N. Rohith'.

**RUBRICS FOR CALCULATION:**

| <b>Criteria</b>                           | <b>Weightage</b> | <b>Level 4 – Exemplary</b>                                                                           | <b>Level 3 – Proficient</b>                                             | <b>Level 2 – Developing</b>                    | <b>Level 1 – Beginning</b>                           |
|-------------------------------------------|------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|------------------------------------------------|------------------------------------------------------|
| <b>Selection of Studies/Parameters</b>    | <b>20%</b>       | <b>Selects highly relevant and diverse studies with well-defined comparison parameters</b>           | <b>Selects relevant studies with minor gaps in parameter definition</b> | <b>Limited or partially relevant selection</b> | <b>Poor or inappropriate selection of studies</b>    |
| <b>Comparison &amp; Differentiation</b>   | <b>25%</b>       | <b>Clearly compares multiple aspects (methods, results, limitations) with strong differentiation</b> | <b>Good comparison with minor gaps</b>                                  | <b>Basic comparison with limited depth</b>     | <b>No clear comparison; mostly descriptive</b>       |
| <b>Critical Evaluation</b>                | <b>25%</b>       | <b>Critically evaluates strengths, weaknesses, and implications with strong justification</b>        | <b>Provides evaluation with some justification</b>                      | <b>Limited evaluation; lacks depth</b>         | <b>No critical evaluation provided</b>               |
| <b>Synthesis &amp; Insight Generation</b> | <b>20%</b>       | <b>Generates meaningful insights and conclusions from comparisons</b>                                | <b>Some insights derived from comparison</b>                            | <b>Limited or unclear insights</b>             | <b>No insights generated</b>                         |
| <b>Clarity</b>                            | <b>10%</b>       | <b>Information is clearly presented (tables/figures) with logical structure</b>                      | <b>Generally clear with minor issues</b>                                | <b>Presentation lacks clarity or structure</b> | <b>Poor presentation and difficult to understand</b> |

# Comparative Analysis

## Q1. CNN vs Fully Connected Neural Network

| Feature                    | CNN                                                  | Fully Connected Neural Network                         |
|----------------------------|------------------------------------------------------|--------------------------------------------------------|
| Architecture               | Uses convolution, pooling and fully connected layers | Mainly uses fully connected layers                     |
| Motivation                 | Designed specifically for image and spatial data     | General-purpose data processing                        |
| Spatial feature extraction | Preserves spatial relationships between pixels       | Spatial relationships are mostly lost after flattening |
| Parameters                 | Fewer because of parameter sharing                   | Very large number of parameters                        |
| Computational complexity   | Lower for image processing                           | Higher for high-resolution images                      |
| Image handling             | Highly suitable                                      | Less suitable                                          |
| Feature learning           | Automatically learns edges, textures and shapes      | Learns features from flattened pixels                  |
| Translation handling       | Can detect features at different positions           | Poorer handling of positional changes                  |

### Role of CNN Layers

- **Convolutional layers:** Extract important visual features.
- **Filters:** Detect edges, textures, shapes and other patterns.
- **Pooling layers:** Reduce feature-map size and computation while retaining important information.
- **Fully connected layers:** Use extracted features for final classification.

### Conclusion

CNNs are more suitable for image classification because they preserve spatial information, use fewer parameters through parameter sharing, and automatically learn hierarchical visual features.

## Q2. Different CNN Layers and Filters

| Layer           | Main Function        | Spatial Information                   | Computational Requirement       |
|-----------------|----------------------|---------------------------------------|---------------------------------|
| Convolution     | Feature extraction   | Preserves local spatial relationships | Moderate                        |
| Activation      | Adds non-linearity   | Preserves feature-map structure       | Low                             |
| Pooling         | Reduces dimensions   | Retains important features            | Low                             |
| Fully Connected | Final classification | Spatial information mostly flattened  | High if feature vector is large |

### Early-Layer Filters

Early filters learn **simple features**, such as:

- Edges
- Lines
- Corners
- Basic textures

### Deeper-Layer Filters

Deeper filters combine simple features to learn:

- Shapes
- Complex textures
- Object parts
- Defect patterns

For example, in industrial defect detection:

**Edges → Surface texture → Shape → Crack/Scratch/Defect**

### Conclusion

The layers work together hierarchically. Convolution extracts features, activation introduces non-linearity, pooling reduces complexity, and fully connected layers perform classification. Deeper

filters provide increasingly meaningful information, improving defect detection.

### Q3. Parameter Sharing vs Independent Weights

| Feature                         | CNN Parameter Sharing                 | Fully Connected Independent Weights         |
|---------------------------------|---------------------------------------|---------------------------------------------|
| <b>Weights</b>                  | Same filter reused across image       | Different weights for different connections |
| <b>Number of parameters</b>     | Much smaller                          | Very large                                  |
| <b>Memory</b>                   | Lower                                 | Higher                                      |
| <b>Computational complexity</b> | Lower                                 | Higher for large images                     |
| <b>Feature detection</b>        | Same feature can be detected anywhere | Position-dependent                          |
| <b>Learning efficiency</b>      | High                                  | Lower                                       |
| <b>Image processing</b>         | Highly efficient                      | Less efficient                              |

#### Example

Suppose a CNN uses a **3×3 filter**.

It has only:

**9 weights + 1 bias**

The same filter can scan the entire image.

In contrast, a fully connected network needs separate weights for connections between every input pixel and neuron.

#### Importance of Parameter Sharing

Parameter sharing allows a CNN to:

- Reduce the number of trainable parameters.

- Reduce memory usage.
- Reduce computation.
- Learn features efficiently.
- Detect the same feature at different image locations.

It provides **translation-related robustness**, meaning a feature such as an edge or defect can be recognized even when its position changes.

## Conclusion

Parameter sharing is essential for large-scale image processing because thousands of high-resolution images can be processed with significantly fewer parameters and lower memory requirements.

## Q4. AlexNet vs ResNet

| Feature               | AlexNet                      | ResNet                                           |
|-----------------------|------------------------------|--------------------------------------------------|
| Depth                 | Relatively shallow           | Much deeper                                      |
| Layer organization    | Conventional sequential CNN  | Uses residual/skip connections                   |
| Convolutional filters | Learns hierarchical features | Learns more complex hierarchical features        |
| Parameters            | About 60 million             | Depends on version                               |
| Feature extraction    | Good                         | Excellent for complex tasks                      |
| Training difficulty   | Relatively easier            | Deep networks made easier using skip connections |
| Vanishing gradient    | More susceptible             | Strongly reduced                                 |
| Computational cost    | Generally lower              | Depends on ResNet version                        |
| Performance           | Good                         | Generally better on complex recognition tasks    |

## AlexNet

AlexNet introduced important CNN techniques such as:

- ReLU activation
- Convolutional feature extraction
- Pooling
- Dropout

It was highly successful for its time but is relatively shallow compared with modern architectures.

## ResNet

ResNet uses **residual/skip connections** that allow information and gradients to pass directly through multiple layers.

This makes it possible to train much deeper networks.

## Recommendation

**ResNet is more suitable for complex image recognition.**

Reasons:

1. Better feature-learning capability.
2. Supports much deeper networks.
3. Reduces the vanishing-gradient problem.
4. Usually provides better classification performance.
5. Can learn complex visual patterns.

A smaller version such as **ResNet-18** or **ResNet-34** can be selected when computational resources are limited.

## Q5. Regularized CNN vs Non-Regularized CNN

| Feature            | Without Regularization | With Regularization    |
|--------------------|------------------------|------------------------|
| Training accuracy  | Very high              | Usually slightly lower |
| Overfitting        | High                   | Reduced                |
| Generalization     | Poor                   | Better                 |
| Unseen images      | Poor performance       | Better performance     |
| Training stability | May be unstable        | More stable            |



|                                   |        |                 |
|-----------------------------------|--------|-----------------|
| <b>Computational requirements</b> | Lower  | Slightly higher |
| <b>Real-world performance</b>     | Poorer | Better          |

## 1. Dropout

Dropout randomly disables some neurons during training.

Example:

**Dropout = 0.5**

It prevents the model from depending too heavily on particular neurons and reduces overfitting.

## 2. Data Augmentation

New training variations can be generated using:

- Rotation
- Flipping
- Scaling
- Cropping
- Brightness changes
- Contrast changes

For leaf disease classification, augmentation helps the CNN recognize diseases under different image conditions.

## 3. Batch Normalization

Batch normalization normalizes intermediate activations.

Benefits:

- Improves training stability.
- Can speed up convergence.
- Reduces sensitivity to initialization.
- Provides some regularization.

## Recommended Approach

For plant disease classification, use **all three techniques together**:

**Data Augmentation + Batch Normalization + Dropout**

A suitable flow is:

**Input → Convolution → Batch Normalization → ReLU → Pooling → Convolution → Batch Normalization → ReLU → Pooling → Dropout → Fully Connected → Output**

## **Conclusion**

A non-regularized CNN may achieve very high training accuracy but perform poorly on unseen leaf images. A regularized CNN provides better generalization and is therefore more reliable for real-world plant disease classification.